# Cortex v1.0
# Comparative evaluation

Seven-category short evaluation - NVIDIA DGX Spark - BF16

## Completion report

| Run | Completed jobs | Observed wall time |
|---|---|---|
| runs-short-all-v1 | 53 / 53 | 2.34 hours |

## Interpretation

These are small exploratory samples, not a full benchmark. No overall winner, statistical significance, production readiness or training-data independence is established. Raw-completion results and native-assistant results are separate comparison tracks.

Short exploratory evaluation, not a full benchmark. Standard tests: first 10 test examples per subtask; KMMLU/MMLU sample 3 domains. Chat/tools use new synthetic constraint-based diagnostics. 64K is Cortex-only. No training-overlap audit established.

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## Audit notes requiring review

Keural-Cortex-8B-v1.0 / GSM8K: Generation override observed: max_new_tokens=2048. Requested harness cap was 512; generation-budget comparability needs review.

Qwen3-8B-Base / GSM8K: Generation override observed: max_new_tokens=2048. Requested harness cap was 512; generation-budget comparability needs review.

Keural-Cortex-8B-v1.0 / GSM8K: Generation override observed: max_new_tokens=2048. Requested harness cap was 512; generation-budget comparability needs review.

## Models and fixed revisions

mkd-hossain/Keural-Cortex-8B-v1.0

Revision: a592a43fe940299b5a025e5b719459cda97ae1e6

Source: https://huggingface.co/mkd-hossain/Keural-Cortex-8B-v1.0

Qwen/Qwen3-8B-Base

Revision: 49e3418fbbbca6ecbdf9608b4d22e5a407081db4

Source: https://huggingface.co/Qwen/Qwen3-8B-Base

Qwen/Qwen3-8B

Revision: b968826d9c46dd6066d109eabc6255188de91218

Source: https://huggingface.co/Qwen/Qwen3-8B

## Protocol

All models use BF16 on the same NVIDIA GB10. Standard task seeds: 0,1234,1234,1234. Standard tasks take the first 10 test examples per selected subtask. This fixed convenience sample is not a representative full-suite estimate.

KMMLU samples computer science, Korean history and economics. MMLU samples computer security, world religions and high-school geography. HAE-RAE samples its child tasks. Five shots are used for KMMLU, HAE-RAE, MMLU and GSM8K; zero for the other standard tasks.

Matched completion compares Cortex to Qwen3-8B-Base without a chat template. Native assistant compares Cortex to Qwen3-8B using their respective chat templates, with thinking disabled for standard and operational cases.

The synthetic assistant diagnostic contains 14 bilingual chat constraints, 13 mocked tool cases and 6 arithmetic/word-problem cases. Thinking math uses seeded temperature 0.6, top-p 0.95, top-k 20, with a 1536-token output budget. Other custom cases allow 256 output tokens. Truncated responses count as failures.

Tool evaluation checks tool selection, argument schema and specified content, batched search, mocked multi-step actions, result grounding and safe failed-lookup behavior. No external tools or real email actions are executed. These are newly written constraint tests, not a standardized tool benchmark.

Long-context probes use two deterministic synthetic documents per 4K/16K/32K context budget for both assistants. They combine key retrieval, addition across records and source labels. Input lengths reserve output space. The 64K extension is Cortex-only. These are not RULER or LongBench results.

The requested GSM8K harness generation cap is 512 tokens. Runtime override warnings, where detected, are listed explicitly. Differences affected by such warnings are withheld pending review of the effective generation budgets.

Raw per-case samples and task fingerprints are preserved on the evaluation host. This PDF summarizes their aggregate metrics; training overlap and dataset decontamination have not been established.

# Matched completion

Cortex v1.0 vs Qwen3-8B-Base. Differences are percentage points; no significance claim.

| Benchmark / subtask | Cortex v1.0 | Qwen3-8B-Base | Difference | Metric |
|---|---|---|---|---|
| KMMLU / computer science | 70.0%<br>n=10 | 80.0%<br>n=10 | -10.0 pp | Accuracy |
| KMMLU / korean history | 40.0%<br>n=10 | 50.0%<br>n=10 | -10.0 pp | Accuracy |
| KMMLU / economics | 60.0%<br>n=10 | 70.0%<br>n=10 | -10.0 pp | Accuracy |
| HAE-RAE / general knowledge | 80.0%<br>n=10 | 90.0%<br>n=10 | -10.0 pp | Normalized accuracy |
| HAE-RAE / history | 80.0%<br>n=10 | 80.0%<br>n=10 | +0.0 pp | Normalized accuracy |
| HAE-RAE / loan word | 100.0%<br>n=10 | 100.0%<br>n=10 | +0.0 pp | Normalized accuracy |
| HAE-RAE / rare word | 70.0%<br>n=10 | 60.0%<br>n=10 | +10.0 pp | Normalized accuracy |
| HAE-RAE / standard nomenclature | 80.0%<br>n=10 | 80.0%<br>n=10 | +0.0 pp | Normalized accuracy |
| kobest boolq | 50.0%<br>n=10 | 60.0%<br>n=10 | -10.0 pp | Accuracy |
| kobest copa | 70.0%<br>n=10 | 70.0%<br>n=10 | +0.0 pp | Accuracy |
| kobest hellaswag | 80.0%<br>n=10 | 70.0%<br>n=10 | +10.0 pp | Normalized accuracy |
| kobest sentineg | 90.0%<br>n=10 | 70.0%<br>n=10 | +20.0 pp | Accuracy |
| MMLU / computer security | 90.0%<br>n=10 | 90.0%<br>n=10 | +0.0 pp | Accuracy |
| MMLU / world religions | 90.0%<br>n=10 | 90.0%<br>n=10 | +0.0 pp | Accuracy |
| MMLU / high school geography | 90.0%<br>n=10 | 80.0%<br>n=10 | +10.0 pp | Accuracy |
| hellaswag | 50.0%<br>n=10 | 50.0%<br>n=10 | +0.0 pp | Normalized accuracy |
| arc challenge | 40.0%<br>n=10 | 60.0%<br>n=10 | -20.0 pp | Normalized accuracy |
| winogrande | 80.0%<br>n=10 | 80.0%<br>n=10 | +0.0 pp | Accuracy |
| gsm8k | 60.0%<br>n=10 | 80.0%<br>n=10 | Review budget | Exact match (strict) |

# Native assistant

Cortex v1.0 vs Qwen3-8B. Differences are percentage points; no significance claim.

| Benchmark / subtask | Cortex v1.0 | Qwen3-8B | Difference | Metric |
|---|---|---|---|---|
| Chat | 71.4%<br>n=14 | 85.7%<br>n=14 | -14.3 pp | Constraint pass rate |
| Math reasoning | 33.3%<br>n=6 | 83.3%<br>n=6 | -50.0 pp | Constraint pass rate |
| Tool calling | 30.8%<br>n=13 | 92.3%<br>n=13 | -61.5 pp | Constraint pass rate |
| KMMLU / computer science | 60.0%<br>n=10 | 40.0%<br>n=10 | +20.0 pp | Accuracy |
| KMMLU / korean history | 10.0%<br>n=10 | 30.0%<br>n=10 | -20.0 pp | Accuracy |
| KMMLU / economics | 40.0%<br>n=10 | 50.0%<br>n=10 | -10.0 pp | Accuracy |
| HAE-RAE / general knowledge | 70.0%<br>n=10 | 90.0%<br>n=10 | -20.0 pp | Normalized accuracy |
| HAE-RAE / history | 60.0%<br>n=10 | 50.0%<br>n=10 | +10.0 pp | Normalized accuracy |
| HAE-RAE / loan word | 100.0%<br>n=10 | 100.0%<br>n=10 | +0.0 pp | Normalized accuracy |
| HAE-RAE / rare word | 70.0%<br>n=10 | 40.0%<br>n=10 | +30.0 pp | Normalized accuracy |
| HAE-RAE / standard nomenclature | 80.0%<br>n=10 | 60.0%<br>n=10 | +20.0 pp | Normalized accuracy |
| kobest boolq | 50.0%<br>n=10 | 60.0%<br>n=10 | -10.0 pp | Accuracy |
| kobest copa | 70.0%<br>n=10 | 50.0%<br>n=10 | +20.0 pp | Accuracy |
| kobest hellaswag | 70.0%<br>n=10 | 40.0%<br>n=10 | +30.0 pp | Normalized accuracy |
| kobest sentineg | 60.0%<br>n=10 | 60.0%<br>n=10 | +0.0 pp | Accuracy |
| MMLU / computer security | 90.0%<br>n=10 | 70.0%<br>n=10 | +20.0 pp | Accuracy |
| MMLU / world religions | 100.0%<br>n=10 | 80.0%<br>n=10 | +20.0 pp | Accuracy |
| MMLU / high school geography | 70.0%<br>n=10 | 80.0%<br>n=10 | -10.0 pp | Accuracy |
| hellaswag | 30.0%<br>n=10 | 50.0%<br>n=10 | -20.0 pp | Normalized accuracy |
| arc challenge | 40.0%<br>n=10 | 60.0%<br>n=10 | -20.0 pp | Normalized accuracy |
| winogrande | 60.0%<br>n=10 | 70.0%<br>n=10 | -10.0 pp | Accuracy |
| gsm8k | 70.0%<br>n=10 | 70.0%<br>n=10 | Review budget | Exact match (strict) |

Synthetic pass rates and standardized task accuracies measure different things. Do not average them into one overall score.

# Long context

Cortex v1.0 vs Qwen3-8B. Differences are percentage points; no significance claim.

| Benchmark / subtask | Cortex v1.0 | Qwen3-8B | Difference | Metric |
|---|---|---|---|---|
| Long context | 0.0%<br>n=2 | 100.0%<br>n=2 | -100.0 pp | Constraint pass rate |
| Long context | 0.0%<br>n=2 | 100.0%<br>n=2 | -100.0 pp | Constraint pass rate |
| Long context | 0.0%<br>n=2 | 100.0%<br>n=2 | -100.0 pp | Constraint pass rate |
| Long context<br>Cortex-only extension | 0.0%<br>n=2 | Not in matched test | - | Constraint pass rate |

Context labels are total budgets. Exact input token counts, raw outputs and truncation flags are in the saved custom-sample artifacts. No paired 64K baseline result is claimed.

# Job inventory

| Model / comparison track | Task | State | Sampling / settings |
|---|---|---|---|
| Keural-Cortex-8B-v1.0<br>Native chat; mocked tools | Chat + tools + thinking math | completed | 14 chat / 13 tools / 6 thinking math cases |
| Keural-Cortex-8B-v1.0<br>Matched raw completion | KMMLU (3 domains) | completed | n=10 per subtask; 5-shot |
| Keural-Cortex-8B-v1.0<br>Matched raw completion | HAE-RAE | completed | n=10 per subtask; 5-shot |
| Keural-Cortex-8B-v1.0<br>Matched raw completion | KoBEST BoolQ | completed | n=10 per subtask; 0-shot |
| Keural-Cortex-8B-v1.0<br>Matched raw completion | KoBEST COPA | completed | n=10 per subtask; 0-shot |
| Keural-Cortex-8B-v1.0<br>Matched raw completion | KoBEST HellaSwag | completed | n=10 per subtask; 0-shot |
| Keural-Cortex-8B-v1.0<br>Matched raw completion | KoBEST SentiNeg | completed | n=10 per subtask; 0-shot |
| Keural-Cortex-8B-v1.0<br>Matched raw completion | MMLU (3 domains) | completed | n=10 per subtask; 5-shot |
| Keural-Cortex-8B-v1.0<br>Matched raw completion | HellaSwag | completed | n=10 per subtask; 0-shot |
| Keural-Cortex-8B-v1.0<br>Matched raw completion | ARC-Challenge | completed | n=10 per subtask; 0-shot |
| Keural-Cortex-8B-v1.0<br>Matched raw completion | WinoGrande | completed | n=10 per subtask; 0-shot |
| Keural-Cortex-8B-v1.0<br>Matched raw completion | GSM8K | completed | n=10 per subtask; 5-shot |
| Qwen3-8B-Base<br>Matched raw completion | KMMLU (3 domains) | completed | n=10 per subtask; 5-shot |
| Qwen3-8B-Base<br>Matched raw completion | HAE-RAE | completed | n=10 per subtask; 5-shot |
| Qwen3-8B-Base<br>Matched raw completion | KoBEST BoolQ | completed | n=10 per subtask; 0-shot |
| Qwen3-8B-Base<br>Matched raw completion | KoBEST COPA | completed | n=10 per subtask; 0-shot |
| Qwen3-8B-Base<br>Matched raw completion | KoBEST HellaSwag | completed | n=10 per subtask; 0-shot |
| Qwen3-8B-Base<br>Matched raw completion | KoBEST SentiNeg | completed | n=10 per subtask; 0-shot |
| Qwen3-8B-Base<br>Matched raw completion | MMLU (3 domains) | completed | n=10 per subtask; 5-shot |
| Qwen3-8B-Base<br>Matched raw completion | HellaSwag | completed | n=10 per subtask; 0-shot |
| Qwen3-8B-Base<br>Matched raw completion | ARC-Challenge | completed | n=10 per subtask; 0-shot |
| Qwen3-8B-Base<br>Matched raw completion | WinoGrande | completed | n=10 per subtask; 0-shot |
| Qwen3-8B-Base<br>Matched raw completion | GSM8K | completed | n=10 per subtask; 5-shot |

| Model / comparison track | Task | State | Sampling / settings |
|---|---|---|---|
| Keural-Cortex-8B-v1.0<br>Native chat; thinking off | KMMLU (3 domains) | completed | n=10 per subtask; 5-shot |
| Keural-Cortex-8B-v1.0<br>Native chat; thinking off | HAE-RAE | completed | n=10 per subtask; 5-shot |
| Keural-Cortex-8B-v1.0<br>Native chat; thinking off | KoBEST BoolQ | completed | n=10 per subtask; 0-shot |
| Keural-Cortex-8B-v1.0<br>Native chat; thinking off | KoBEST COPA | completed | n=10 per subtask; 0-shot |
| Keural-Cortex-8B-v1.0<br>Native chat; thinking off | KoBEST HellaSwag | completed | n=10 per subtask; 0-shot |
| Keural-Cortex-8B-v1.0<br>Native chat; thinking off | KoBEST SentiNeg | completed | n=10 per subtask; 0-shot |
| Keural-Cortex-8B-v1.0<br>Native chat; thinking off | MMLU (3 domains) | completed | n=10 per subtask; 5-shot |
| Keural-Cortex-8B-v1.0<br>Native chat; thinking off | HellaSwag | completed | n=10 per subtask; 0-shot |
| Keural-Cortex-8B-v1.0<br>Native chat; thinking off | ARC-Challenge | completed | n=10 per subtask; 0-shot |
| Keural-Cortex-8B-v1.0<br>Native chat; thinking off | WinoGrande | completed | n=10 per subtask; 0-shot |
| Keural-Cortex-8B-v1.0<br>Native chat; thinking off | GSM8K | completed | n=10 per subtask; 5-shot |
| Qwen3-8B<br>Native chat; thinking off | KMMLU (3 domains) | completed | n=10 per subtask; 5-shot |
| Qwen3-8B<br>Native chat; thinking off | HAE-RAE | completed | n=10 per subtask; 5-shot |
| Qwen3-8B<br>Native chat; thinking off | KoBEST BoolQ | completed | n=10 per subtask; 0-shot |
| Qwen3-8B<br>Native chat; thinking off | KoBEST COPA | completed | n=10 per subtask; 0-shot |
| Qwen3-8B<br>Native chat; thinking off | KoBEST HellaSwag | completed | n=10 per subtask; 0-shot |
| Qwen3-8B<br>Native chat; thinking off | KoBEST SentiNeg | completed | n=10 per subtask; 0-shot |
| Qwen3-8B<br>Native chat; thinking off | MMLU (3 domains) | completed | n=10 per subtask; 5-shot |
| Qwen3-8B<br>Native chat; thinking off | HellaSwag | completed | n=10 per subtask; 0-shot |
| Qwen3-8B<br>Native chat; thinking off | ARC-Challenge | completed | n=10 per subtask; 0-shot |
| Qwen3-8B<br>Native chat; thinking off | WinoGrande | completed | n=10 per subtask; 0-shot |
| Qwen3-8B<br>Native chat; thinking off | GSM8K | completed | n=10 per subtask; 5-shot |
| Qwen3-8B<br>Native chat; mocked tools | Chat + tools + thinking math | completed | 14 chat / 13 tools / 6 thinking math cases |
| Keural-Cortex-8B-v1.0<br>Native chat; matched context budget | Long context 4K | completed | Context budget 4096 tokens |

| Model / comparison track | Task | State | Sampling / settings |
|---|---|---|---|
| Qwen3-8B<br>Native chat; matched context budget | Long context 4K | completed | Context budget 4096 tokens |
| Keural-Cortex-8B-v1.0<br>Native chat; matched context budget | Long context 16K | completed | Context budget 16384 tokens |
| Qwen3-8B<br>Native chat; matched context budget | Long context 16K | completed | Context budget 16384 tokens |
| Keural-Cortex-8B-v1.0<br>Native chat; matched context budget | Long context 32K | completed | Context budget 32768 tokens |
| Qwen3-8B<br>Native chat; matched context budget | Long context 32K | completed | Context budget 32768 tokens |
| Keural-Cortex-8B-v1.0<br>Cortex-only extension | Long context 64K | completed | Context budget 65536 tokens |